

HO�F–RIESZ: Round 002

Smooth iid residuals and quantitative interfaces

Campaign research memorandum

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Published input checkpoint: `a06178658d1e3d458536ff312ca793947212ec67`. This memorandum records bounded mathematical advances within the fixed smooth iid model. The accompanying Round 002 report records the independent audit verdicts and immutable proof hashes. The singular subcritical and critical missions remain open; no cutoff removal is implicit in a fixed-smooth bound.

1 Model and statement

On the unit torus of volume one let $K = -\nabla g$ for a smooth real even interaction g , and let $b = -\nabla V$ be smooth. For $N \geq 2$, $\beta > 0$,

$$dx_i = \left(b(x_i) + N^{-1} \sum_{j \neq i} K(x_i - x_j) \right) dt + \sqrt{2/\beta} dW_i.$$

Initial positions are iid with smooth positive probability density μ_0 , independently of the Brownian motions. The deterministic reference solves

$$\partial_t \mu = -\operatorname{div}((b + K * \mu)\mu) + \beta^{-1} \Delta \mu, \quad \mu|_{t=0} = \mu_0.$$

Write $\eta = N^{-1} \sum_i \delta_{x_i}$ and $\rho = \eta - \mu$. The centered deleted-label statistic $\mathcal{U}_k$ alternates over all subsets of k slots, filling that subset by distinct particle labels with denominator $N^{|A|}$ and the other slots by μ . All tuples are ordered. Set $P = \mathcal{U}_2/2$ and $\sigma = \sqrt{N} \min(\sqrt{\beta}, 1)$.

Let f be the full backward linearized one-body test and Φ its symmetric zero-terminal pair corrector, with the operators frozen in Round 001. For $r = 4d + 12$ suppose their C^r norms and that of μ are bounded uniformly in $N, \beta, t \in [0, T]$. Put

$$F = C\Phi = \operatorname{Sym}_3[K(x - z) \cdot \nabla_x \Phi(x, y)],$$

where symmetrization averages six permutations. THM-010 establishes

$$\begin{aligned} \mathbb{E}|P[\Phi_0]| &\leq I/N, & \mathbb{E} \int_0^T |\mathcal{U}_3[F_t]| dt &\leq TC_3/N^{3/2}, \\ \mathbb{E}[M_\Phi]_T &\leq 2TH^2/(\beta N^2), & \mathbb{E} \operatorname{TV}[M_f, M_\Phi] &\leq 2TL_f H/(\beta N^{3/2}). \end{aligned}$$

All constants are explicit in the constructor memorandum, equations (2.1)–(2.4), and are independent of N, β . They depend on the fixed kernel, drift, horizon and displayed test norms. In particular the four scaled targets have bounds, respectively,

$$O(N^{-1/2}), \quad O(N^{-1}), \quad O(N^{-1}), \quad O(N^{-1/2}),$$

when the first two are multiplied by σ and the brackets by σ^2 . These are absolute estimates under the actual interacting law.

2 Coupling, moments, and exact deletion

Here is the load-bearing mechanism. Let $q = \lfloor d/2 \rfloor + 2$ and use Fourier characters $e^{2\pi i k \cdot x}$. Define

$$\|\zeta\|_{-q}^2 = \sum_{k \in \mathbb{Z}^d} (1 + 4\pi^2 |k|^2)^{-q} |\hat{\zeta}(k)|^2.$$

The finite constants

$$B^2 = \sum_k (1 + 4\pi^2 |k|^2)^{-q}, \quad D^2 = \sum_k 4\pi^2 |k|^2 (1 + 4\pi^2 |k|^2)^{-q}$$

give $\|\delta_x\|_{-q} = B$ and $\|\delta_x - \delta_y\|_{-q} \leq D|\tilde{x} - \tilde{y}|$. For independent samples of any probability law, expansion of the sixth Hilbert moment shows

$$\left\| N^{-1} \sum_i \delta_{y_i} - \mu \right\|_{L^p(\Omega; H^{-q})} \leq 2 \cdot 3^{7/6} B / \sqrt{N}, \quad 1 \leq p \leq 6.$$

Indeed a singleton index has zero expectation in the six-factor expansion; a surviving list uses at most three labels. The second moment has the sharper bound B^2/N .

Couple the particles to independent nonlinear particles driven by the same initial samples and Brownian motions, with deterministic drift $b + K * \mu_t$. For lifted displacements and their empirical mean D_N , the noises cancel. Writing $\bar{\rho}$ for the independent empirical discrepancy gives

$$D_N(t) \leq \mathcal{K} \int_0^t e^{L(t-s)} \|\bar{\rho}_s\|_{-q} ds, \quad \|\rho_t\|_{-q} \leq \|\bar{\rho}_t\|_{-q} + DD_N(t),$$

where $\mathcal{K} = (\sum_a \|K_a\|_{H^q}^2)^{1/2}$ and $L = d(\|b\|_{C^1} + 2\|K\|_{C^1})$. Minkowski's inequality yields

$$\sup_{t \leq T} \|\rho_t\|_{L^p(\Omega; H^{-q})} \leq \frac{R}{\sqrt{N}}, \quad R = 2 \cdot 3^{7/6} B \left(1 + D\mathcal{K} \frac{e^{LT} - 1}{L} \right),$$

with the fraction interpreted as T at $L = 0$. The supremum is outside the expectation. There is no claim of temperature-uniform supremum-in-time empirical moments.

For symmetric kernels the exact, configuration-wise deletion formulas are

$$\begin{aligned} \mathcal{U}_2[\Phi] &= \rho^{\otimes 2}[\Phi] - N^{-1} \eta[\Phi(x, x)], \\ \mathcal{U}_3[F] &= \rho^{\otimes 3}[F] - 3N^{-1} (\eta \otimes \rho)[F(x, x, y)] + 2N^{-2} \eta[F(x, x, x)]. \end{aligned}$$

The coefficients follow by inclusion-exclusion of label equalities, including $N = 2$. Smooth tensor Sobolev duality bounds the first cubic term by $Q^3 \|F\|_{C^{3q}} \|\rho\|_{-q}^3$, with $Q = (d+1)^{q/2}$. The partial diagonal is bounded by $Q \|F\|_{C^q} \|\rho\|_{-q}$. For the actual source $C\Phi$, the full diagonal is zero since $K(0) = 0$. These observations prove the cubic absolute bound without assuming the evolved law is iid. The pair bound uses the initial second moment and retains its nonzero deletion bias.

The exact pair gradient is $\nabla_{x_i} P = N^{-1} H_i$, where

$$H_i = \int \nabla_1 \Phi(x_i, y) \rho(dy) - N^{-1} \nabla_1 \Phi(x_i, x_i).$$

Sobolev duality in y , uniformly in x_i , gives $|H_i| \leq L_\Phi \|\rho\|_{-q} + D_\Phi/N$. Hence

$$[M_\Phi]_T = \frac{2}{\beta N^2} \sum_i \int_0^T |H_i|^2 dt, \quad [M_f, M_\Phi]_T = \frac{2}{\beta N^2} \sum_i \int_0^T \nabla f(x_i) \cdot H_i dt.$$

These yield the two bracket estimates. Their scaling uses exactly $\sigma^2/\beta = N \min(1, \beta^{-1})$. No signed covariance cancellation is needed. The two lower drift contractions have absolute integral at most $5Td\|K\|_\infty \sup_t \|\Phi_t\|_{C^1}/N$.

3 Qualification of the smooth norms

The separate root lemma records bounds for existing smooth solutions in the frozen model. Write $b_m = \|b\|_{C^m}$, $\kappa_m = \|K\|_{C^m}$ and $U_m = b_m + \kappa_m$. Differentiating the forward density equation and applying the finite-family maximum principle gives

$$\sup_{t \leq T} \|\mu_t\|_{C^m} \leq \|\mu_0\|_{C^m} \exp\{d(2^{m+1} - 1)U_{m+1}T\}.$$

The differentiated transport contributes $d(2^m - 1)U_{m+1}$ and the differentiated reaction contributes $d2^m U_{m+1}$. Nonnegative diffusion introduces no adverse constant. For the one-body backward equation, derivatives in the response operator fall on K ; for $m \geq 1$ its C^m norm is at most $d\kappa_m$. Thus, with $a_m = d(2^m - 1)U_m + d\kappa_m$,

$$\sup_{t \leq T} \|f_t\|_{C^m} \leq e^{a_m T} \|h\|_{C^m}.$$

The audited pair estimate THM-009 then implies, for $m \geq 2$,

$$\sup_{t \leq T} \|\Phi_t\|_{C^m} \leq T e^{\bar{c}_m T} d^{2^{m+1}} \kappa_m e^{a_{m+1} T} \|h\|_{C^{m+1}},$$

where

$$\begin{aligned} \bar{M} &= d\|\mu_0\|_{C^1} e^{3dU_2 T}, \\ \bar{c}_m &= 2d(2^m - 1)(b_m + 3\kappa_m/2) + 2(d\kappa_m + \kappa_0 \bar{M}). \end{aligned}$$

Taking $m = r$ supplies the declared regularity bounds for fixed smooth data, uniformly in N, β . These constants diverge with a singular heat cutoff; this qualification does not discharge a singular estimate.

4 Finite-dimensional Gaussian criterion

Fix a finite list $(t_a, \phi_a)_{a=1}^m$. Let f_a^N denote the full one-body backward test on $[0, t_a]$, with its pair corrector on that interval. Put $a_N = \sigma_N / \sqrt{N}$ and $c_N = \sigma_N^2 / (N\beta_N)$. The deterministic covariance matrices are

$$\begin{aligned} I_N^{ab} &= a_N^2 \text{Cov}_{\mu_0}(f_a^N(0), f_b^N(0)), \\ D_N^{ab} &= 2c_N \int_0^{t_a \wedge t_b} \mu_r^N(\nabla f_a^N(r) \cdot \nabla f_b^N(r)) dr. \end{aligned}$$

THM-011 proves that entrywise convergence $I_N \rightarrow I$, $D_N \rightarrow D$ implies

$$(\sigma_N \rho_{t_a}^N(\phi_a))_{a=1}^m \implies \mathcal{N}_m(0, I + D).$$

The Gaussian may be degenerate. The full proof is the sealed memorandum `ROUND_002_SMOOTH_GAUSSIAN.md`; its independent review is recorded separately in the round report.

The exact corrected identity decomposes the vector as $Z_N = S_N + L_N + R_N$, where S_N is the initial linearized statistic, L_N the leading one-body martingale, and $\|R_N\|_2 \leq CN^{-1/2}$. The independent Fourier residual proof gives the necessary L^2 bounds. Its expectation also proves $|\mathbb{E} \rho_{t_a}^N(\phi_a)| \leq C/N$, so the asserted deterministic mean-field centering is valid at scale σ_N .

For a fixed vector θ , let Q_N^θ be the bracket of $\theta \cdot L_N$. Actual-law empirical control gives

$$\mathbb{E}|Q_N^\theta(T) - \theta^T D_N \theta| \leq C_\theta N^{-1/2}, \quad 0 \leq Q_N^\theta(t) \leq B_\theta,$$

with a deterministic bound. Thus $E_N(t) = \exp(i\theta \cdot L_N(t) + Q_N^\theta(t)/2)$ is a true bounded complex martingale, and $\mathbb{E}[E_N(T) | \mathcal{F}_0] = 1$. For every bounded initial-measurable Y_N ,

$$\left| \mathbb{E}[Y_N e^{i\theta \cdot L_N}] - e^{-\theta^T D_N \theta/2} \mathbb{E}Y_N \right| \leq C_\theta \|Y_N\|_\infty N^{-1/2}.$$

An elementary third-order Taylor bound for the triangular iid initial sum then yields

$$\mathbb{E}e^{i\alpha \cdot S_N + i\theta \cdot L_N} = e^{-\alpha^T I_N \alpha/2 - \theta^T D_N \theta/2} + O_{\alpha, \theta}(N^{-1/2}).$$

Independent Gaussian smoothing and the uniform second moments give weak convergence, including degenerate covariance. This proves independence in the limit, without asserting finite-particle independence.

For general fixed smooth data the full-response energy identity is

$$\frac{d}{dt} \mu_t(f_t^2) = 2\beta_N^{-1} \mu_t(|\nabla f_t|^2) - 2\mu_t(f_t R f_t).$$

The uniform spatial bounds imply $\int_0^{t_a} \mu_r(|\nabla f_a|^2) dr \leq C\beta_N$. Consequently $I_N, D_N \rightarrow 0$ if $\beta_N \rightarrow 0$, and the observable converges to zero in L^2 at this normalization. If $\beta_N \rightarrow \infty$, the leading noise has variance $O(\beta_N^{-1})$; identifying the remaining initial covariance requires convergence of the backward kernels or their covariance.

For free heat, Haar background and a real cosine mode of eigenvalue $\omega_k = 4\pi^2|k|^2$, the exact covariance at times t, s is

$$\frac{\min(\beta_N, 1)}{2} e^{-\omega_k |t-s|/\beta_N}.$$

The initial and noise contributions sum to this formula at every finite N . At time zero, alternating $\beta_N = 1/2$ and $\beta_N = 2$ gives covariance limits $1/4$ and $1/2$. Hence arbitrary oscillating temperatures need not have a single limiting law. No field/path tightness follows from this finite-list theorem.

5 Independent all-order reconstruction

AUD-004 compares an isolated reconstruction with the frozen R1 formulas. Both give the same upward, internal and lower drift terms, with the apparent factor $k(k-1)$ in the new lower kernel becoming k when that kernel is defined as the sum of $k-1$ incident pair terms. Every shared-label bracket also agrees. In a centered representation, a matching of p labels across the two blocks has one distinguished Brownian edge, coefficient $2/(\beta_N N^p)$, and multiplicity

$$p \binom{k}{p} \binom{\ell}{p} p! = k\ell \binom{k-1}{p-1} \binom{\ell-1}{p-1} (p-1)!.$$

Every subset of the p shared quotient slots is then integrated once against μ ; the remaining slots enter the centered statistic. This is an exact factorial-to-centered inversion, valid also for $k > N$. Individual centered terms need not be positive. The finite algebra now has both isolated reconstruction and hostile review passes. Critical effective power counting remains a separate gate.

6 All-order estimates and a sufficient cutoff condition

THM-014 repairs THM-012's missing symmetry qualification. Its partition formula, with full proof and constants in the power-counting memorandum. For a partition π of $[k]$, write $j(\pi)$ for its singleton count and

$$a(\pi) = k - |\pi|, \quad c(\pi) = \prod_{B \in \pi} (|B| - 1)!, \quad \Delta(\pi) = \frac{1}{2} \sum_{|B| \geq 2} (|B| - 2).$$

If Φ_π identifies variables inside each block, then

$$\mathcal{U}_k[\Phi] = \sum_{\pi} (-1)^{a(\pi)} c(\pi) N^{-a(\pi)} \int \Phi_\pi \prod_{|B|=1} \rho(dx_B) \prod_{|B| \geq 2} \eta(dx_B).$$

It holds also when $k > N$. The proof selects injective assignments by the signed cycle-partition weights, then sums occupied-singleton versus background choices to obtain ρ in each remaining singleton slot.

Symmetrization with independent copies and signs gives the iid Hilbert bound

$$\mathbb{E} \|\bar{\rho}_t\|_{-q}^{2m} \leq (2B)^{2m} (2m - 1)!! N^{-m}.$$

The same coupling transfers every fixed moment to the interacting law. With $S = 1 + DK(e^{LT} - 1)/L$ and $R_p = 2BS\sqrt{2\lceil p/2 \rceil}$, it gives $\sup_t \|\rho_t\|_{-q} \leq R_p N^{-1/2}$. Tensor duality in singleton variables therefore proves

$$\|\mathcal{U}_k[\Phi_t]\|_{L^p} \leq \sum_{\pi} c(\pi) (QR_{pj(\pi)})^{j(\pi)} \|\Phi_t\|_{C^{j(\pi)q}} N^{-k/2 - \Delta(\pi)}.$$

At $j = 0$ the moment factor is one. Pair blocks have $\Delta = 0$; larger blocks have a strict extra power. The summed constant $C_{k,p}(N)$ is a finite explicit partition sum, and obeys the conservative fixed-data bound

$$C_{k,p}(N) \leq k! \exp\{2QBS\sqrt{pk + 2}\}.$$

The exact finite sum, rather than this last exponential majorant, is used when the data norms vary with cutoff.

For rooted derivatives use symmetric kernels, replacing any kernel by its averaged symmetrization; this preserves $\mathcal{U}_k$ and contracts every C^m norm. Excluding the root replaces R_p by $R_p + B$ and gives constants $D_{k-1,2}(N)$ defined by the same partition sum. In particular, if $B'_k = \sup_t \|\Phi_{k,t}\|_{C^{(k-1)q+1}}$,

$$\mathbb{E} \text{TV}[M_k, M_\ell] \leq \frac{2Tdk\ell}{\beta_N} B'_k B'_\ell D_{k-1,2}(N) D_{\ell-1,2}(N) N^{-(k+\ell)/2}.$$

The scaled power is $N^{1-(k+\ell)/2} \min(1, \beta_N^{-1})$. The detailed generator table also retains the derivative cost of every upward, internal, single-lowering and double-lowering term. A double contraction is of the same raw order as $\mathcal{U}_k$; it is not a new gain at its own level. Infinite summation requires the displayed order-dependent norms and coefficients to be summable, including all drift and cross-bracket terms. A separately stated lemma with geometric kernel-norm growth and coefficients $1/k!$ proves statistical and martingale tail control; those extra hypotheses are not asserted for the actual hierarchy.

For the frozen positive-power Riesz heat sequence, $0 < s < d$, the report proves explicit constants with

$$\|K_\varepsilon\|_{C^m} \leq L_m \varepsilon^{-(s+m+1)/2}, \quad \|K_\varepsilon\|_{C^1} \geq c \varepsilon^{-(s+2)/2}.$$

The gamma factors in L_m and the lattice lower bound are displayed in the proof. Thus the particular coupling constant grows exponentially in $\varepsilon^{-(s+2)/2}$ on a fixed positive horizon. For the regularized pair problem, a sufficient joint-limit condition is

$$\varepsilon_N^{-(s+2)/2} + \log \left(1 + \sup_t \|\Phi_t^{N, \varepsilon_N}\|_{C^{3q+1}} + \sup_t \|f_t^{N, \varepsilon_N}\|_{C^1} \right) = o(\log N).$$

All four regularized residual targets then vanish, uniformly over positive temperatures. If those norm envelopes grow polynomially, a sufficient choice is $\varepsilon_N = (\log N)^{-\gamma}$ with $0 < \gamma < 2/(s+2)$. More generally finite uniform envelopes at each fixed cutoff permit a sufficiently slow diagonal choice. The fixed-data qualification supplies such finite envelopes for the existing smooth family; it does not imply polynomial cutoff growth.

The missing comparison is a different assertion, for example

$$\sigma_N \left\langle \phi, (\eta_T^{\text{sing}} - \mu_T^{\text{sing}}) - (\eta_T^{\varepsilon_N} - \mu_T^{\varepsilon_N}) \right\rangle \longrightarrow 0$$

in a specified mode, with singular well-posedness justified. A cutoff slow enough for these derivative estimates may smooth the microscopic scales needed for that comparison. No comparison or critical summation is claimed.

7 Diffusivity continuity and identified smooth limits

The separate THM-013 proof constructs the smooth inviscid reference by the characteristic fixed point

$$F_t(x) = x + \int_0^t \left[b(F_s(x)) + \int K(F_s(x) - F_s(y)) \mu_0(y) dy \right] ds.$$

On periodic displacements its right side is Lipschitz with constant $L_b + 2L_K$. Bounded speed allows continuation on every finite horizon. The resulting smooth flow has positive Jacobian determinant and pushes μ_0 to the unique smooth inviscid solution. Smooth positive-diffusivity mean-field existence remains the declared model input. The full inviscid backward equation is constructed by a Volterra series for transport plus the bounded response; its n th term has the displayed factorial denominator.

For every fixed m the proof establishes

$$\begin{aligned} \sup_{t \leq T} \|\mu_t^\nu - \mu_t^{\nu'}\|_{C^m} &\leq C_{\mu, m} |\nu - \nu'|, \\ \max_a \sup_{t \leq t_a} \|f_a^\nu(t) - f_a^{\nu'}(t)\|_{C^m} &\leq C_{f, m} |\nu - \nu'|, \end{aligned}$$

with explicit fixed-data constants and no inverse diffusivity, including $\nu = 0$. Two additional derivatives of the density or backward test are required by the Laplacian forcing. In the backward difference, both the velocity difference and

$$(R_{\mu^\nu} - R_{\mu^{\nu'}})f(y) = \int K(z - y) \cdot \nabla f(z) (\mu^\nu - \mu^{\nu'})(z) dz$$

are retained. Output derivatives hit K , so this last term needs only the supremum norm of the density difference. Maximum-principle commutators and Gronwall prove the stated bounds; there is no unproved parabolic smoothing estimate at $\nu = 0$.

It follows that the covariance matrices in THM-011 converge for every finite positive limiting inverse temperature. Their limits are exactly the formulas already displayed, evaluated at the limiting reference and full backward kernels. If $\beta_N \rightarrow \infty$, then $\nu_N \rightarrow 0$, and

$$I_N^{ab} \longrightarrow \text{Cov}_{\mu_0}(f_a^0(0), f_b^0(0)), \quad D_N \longrightarrow 0.$$

Combined with the high-temperature zero limit, these results identify every physical-temperature convergent subsequence in the fixed-smooth finite-list model. For $0 < s < d$, a critical-labelled sequence $\beta_N \sim \lambda N^{1-s/d}$ tends to infinity. This gives the inviscid initial Gaussian for a *fixed smooth* interaction only; it is not the singular critical law.

Exact checks include free heat, zero/equal/different terminal times, both signs of a one-mode interaction and its removable resonance. The full one-mode response yields

$$f_a^\nu(t, x) = e^{-(2\pi)^2(\nu+a/2)(t_a-t)} \cos(2\pi x).$$

A high-frequency free test also shows why a uniform Lipschitz estimate cannot depend solely on the terminal C^m norm without extra derivatives.

8 Law-class obstruction

For the smooth circle interaction $g(x) = a \cos(2\pi x)$ with $a < 0$, canonical Gibbs density is proportional to $\exp(\beta|a|N|\eta(e_1)|^2/2)$. Translation invariance makes the one-point marginal uniform. If every particle lies in a sufficiently short fixed arc of length $p_\delta > 0$, then $|\eta(e_1)|^2 \geq 1 - \delta/2$. Its contribution to the partition function gives

$$\mathbb{P}_{\text{Gibbs}}(|\eta(e_1)|^2 \geq 1 - \delta) \leq p_\delta^{-N} e^{-\beta|a|N\delta/4}.$$

For every $\beta_N \rightarrow \infty$, the pair mode $\mathcal{U}_2[\cos(2\pi(x-y))] = |\eta(e_1)|^2 - 1/N$ tends to one in probability. Thus the iid moment estimates cannot be transferred to arbitrary smooth Gibbs preparation, even with exact first-marginal centering. This hostile- reviewed obstruction is outside the iid theorem and is not a counterexample to a positive-Riesz singular statement.

9 Scope and next load-bearing interface

The old condition $\beta_N N^{2s/d-1} \rightarrow 0$, full microscopic subcriticality $\lambda_N = \beta_N N^{s/d-1} \rightarrow 0$, and criticality $\lambda_N \rightarrow \lambda \in (0, \infty)$ remain distinct. No Riesz exponent occurs in the fixed-smooth coupling argument. Its higher-order and cutoff constants must be controlled before they can decide singular critical survival, truncation, or resummation. In particular an exponential in $T\|\nabla K_\varepsilon\|_\infty$ is not an effective power of λ_N .

No assertion about Gibbs preparation, the physical gap process, exact-marginal centering, source covariance normalization, or field/path tightness is added by this memorandum. Those interfaces retain their separate campaign records.